

$$T = \frac{2\pi}{\omega}$$

$$\Delta T = \left| \frac{dT}{d\omega} \right| \Delta \omega$$

$$\frac{dT}{d\omega} = \frac{d}{d\omega} \left(\frac{2\pi}{\omega} \right) = -\frac{2\pi}{\omega^2}$$

$$\left| \frac{dT}{d\omega} \right| = \frac{2\pi}{\omega^2}$$

$$\Delta T = \frac{2\pi}{\omega^2} \Delta \omega$$